Reviewer Pack — Minimal Rank of Algebraic Automorphism Groups vs Monotone k-…

Entry a42e66e6e38c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-25 22:11:46 UTC

Minimal Rank of Algebraic Automorphism Groups vs Monotone k-CLIQUE Circuit Size

Entry ID: `a42e66e6e38c` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-25 22:11:46 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Automorphism Groups) **Field B** (complexity object): Complexity Theory: Monotone Circuit Complexity for k-CLIQUE

Statement:

> ['For any n-vertex k-CLIQUE problem, the minimal rank of the algebraic automorphism group of its defining graph is at most polynomial in n.', 'There exists a constant c such that for all n-vertex k-CLIQUE graphs G, the minimal rank of Aut(G) ≤ cn.']

Rationale (proposer's reasoning):

> ['Algebraic automorphism groups capture symmetry properties of a graph, and understanding these symmetries could provide insights into the complexity of computational problems.', "Monotone k-CLIQUE is a hard problem, and if there exists a strong connection between its automorphism group's rank and the circuit size required to solve it, this might lead to new algorithmic techniques or complexity lower bounds."]

Taxonomy category: `MONOTONE_CLIQUE` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `6a84e1d003c54255`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if for all n-vertex k-CLIQUE graphs, the minimal rank of Aut(G) is at most cn, where c is a constant and n is the number of vertices.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal Rank Automorphism Groups AND Monotone k-CLIQUE Circuit Complexity - Polynomial Bound Algebraic Geometry Automorphism Groups ON k-CLIQUE Circuit Size - Aut(G) <= cn Constant for k-CLIQUE Graphs AND Algebraic Geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=0, elapsed=0.0s

5.1 Generated Python source

```python
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_clique_graph(n, k):
        if n < k:
            return None
        vertices = list(range(n))
        edges = []
        for i in range(k):
            for j in range(i + 1, k):
                edges.append((vertices[i], vertices[j]))
        for _ in range(k, n):
            u = random.choice(vertices)
            v = random.choice(vertices)
            if (u, v) not in edges and (v, u) not in edges:
                edges.append((u, v))
        return edges

    def is_connected(graph, n):
        visited = [False] * n
        stack = [0]
        while stack:
            node = stack.pop()
            if not visited[node]:
                visited[node] = True
                for neighbor in graph:
                    if neighbor[0] == node and not visited[neighbor[1]]:
                        stack.append(neighbor[1])
                    elif neighbor[1] == node and not visited[neighbor[0]]:
                        stack.append(neighbor[0])
```

```
    return all(visited)

def find_automorphisms(graph, n):
    if not is_connected(graph, n):
        return 0
    automorphisms = set()
    for perm in itertools.permutations(range(n)):
        new_graph = [(perm[u], perm[v]) for u, v in graph]
        if sorted(new_graph) == sorted(graph):
            automorphisms.add(tuple(perm))
    return len(automorphisms)

def min_rank(automorphisms):
    if not automorphisms:
        return 0
    n = len(automorphisms)
    adjacency_matrix = [[0] * n for _ in range(n)]
    for i, perm in enumerate(automorphisms):
        for j, p in enumerate(perm):
            adjacency_matrix[i][j] = adjacency_matrix[j][i] = (p == i)
    rank = 0
    for row in adjacency_matrix:
        if any(row):
            rank += 1
            for i in range(n):
                if row[i]:
                    for j in range(i + 1, n):
                        if adjacency_matrix[j][i]:
                            adjacency_matrix[j][i] = False
    return rank

results = []
for n in [10, 20, 30, 40]:
    graph = generate_k_clique_graph(n, k=5)
    if graph is None:
        continue
    automorphisms = find_automorphisms(graph, n)
    rank = min_rank(automorphisms)
    results.append(rank)

if not results:
    return {
        "metric_name": "min_rank",
        "metric_value": 0,
        "instances_tested": 0,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

mean = sum(results) / len(results)
std = (sum((x - mean) ** 2 for x in results) / len(results)) ** 0.5
support_fraction = Fraction(sum(1 for r in results if r <= 3 * n), len(results))

return {
    "metric_name": "min_rank",
    "metric_value": mean,
```

```
        "instances_tested": len(results),
        "conjecture_holds": support_fraction >= Fraction(8, 10),
        "counterexample": "" if support_fraction >= Fraction(8, 10) else f"mean={mean}, std={std}'
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        if not result["conjecture_holds"]:
            break
        results.append(result["metric_value"])

    if len(results) == len(seeds):
        mean = sum(results) / len(results)
        std = (sum((x - mean) ** 2 for x in results) / len(results)) ** 0.5
        support_fraction = Fraction(len(results), len(seeds))
        print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")
    elif result["counterexample"]:
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed
    else:
        print("RESULT: INCONCLUSIVE reason=budget_exceeded n_tested=30")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
ture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'min_rank', 'metric_value': 0.0, 'instances_tested': 4, 'conjecture_holds': T
TRIAL: {'metric_name': 'min_rank', 'metric_value': 0.0, 'instances_tested': 4, 'conjecture_holds': T
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TRIAL: {'metric_name': 'min_rank', 'metric_value': 0.0, 'instances_tested': 4, 'conjecture_holds': T
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

The empirical test has only tested n ≤ 15 instances, which is insufficient to validate a conjecture that concerns the behavior of the minimal rank for all n-vertex k-CLIQUE graphs. The metric does not necessarily scale trivially with n, and testing such a small range may not capture the true complexity of the problem.

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Safety rail: critic_challenge | original judge: The test results indicate that for all tested instances (n ≤ 15), the conjecture holds true with a mean of 0.0 and standard deviation of 0.0, which me | next: Further testing is required to validate the conjecture for larger values of n. It would be beneficial to test up to n = 40 as per the pre-registered criterion.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11838
2	propose	ollama_remote	glm4:latest	0	0	9740
3	preregistration	ollama_remote	glm4:latest	0	0	6132
4	novelty	ollama_remote	glm4:latest	0	0	4701
5	novelty	ollama_remote	glm4:latest	0	0	5387
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13453
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10101
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9738
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11787
10	critic	ollama_remote	glm4:latest	0	0	9661
11	judge	ollama_remote	glm4:latest	0	0	5757

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 98296 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/a42e66e6e38c.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/a42e66e6e38c.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/a42e66e6e38c.tar.gz` (if generated)